

Effect of Grain Contact Elastic Properties of Unconsolidated Sandstones Under Confining Stress: Insights from Discrete Element Method

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ABSTRACT

Unconsolidated sandstones typically exhibit high porosities and permeabilities. Their large storage and flow capacities make them particularly appealing for geological CO₂ sequestration and wastewater disposal while hosting numerous heavy oil reservoirs concurrently, and they are of great interest in contemporary research (Pothana et al., 2024). The geophysical/geomechanical characterization of such unconsolidated sandstone formations is necessary for safe geoenvironmental applications. Particular interest are the grain contact properties are the dominating factors that control these rocks' elasticity and strength. The Hertz-Mindlin contact theory tends to overestimate velocities and underestimate Poisson's ratios when applied to unconsolidated sands (Zimmer et al., 2002). Under confining loading conditions, these rocks display relatively higher compliance than consolidated rocks, which is attributed to the higher grain contact compliance of the former (Bachrach & Avseth, 2008; Sayers, 2021). Although it is clearly understood that the shear stiffness at grain contacts is markedly lower than the normal stiffness, how the relative values affect the effective compliance rock is ongoing research (Sayers & Dasgupta, 2024). In addition, the variation of the contact area between the grains under confining pressure is rarely considered in rock physics modelling of these rocks.

The applicability of traditional homogenization models, rooted in continuum mechanics, is restricted when dealing with granular materials. Granular rocks inherently display heterogeneity and a non-uniform force networks (Jaeger & Nagel, 1992). Furthermore, granular materials manifest an arching mechanism, resulting in stress and strain localization that are difficult to model using continuum mechanics (Goldenberg & Goldhirsch, 2002). The application of the Discrete Element Method (DEM) emerges as an efficient numerical tool developed based on the physics governing the interactions among rigid grains. This method uses contact laws, functions of normal and shear stiffness at the contact level, among other parameters. DEM efficiently incorporates translational and rotational degrees of freedom. Compliance of unconsolidated rocks under confining pressure can be analyzed while considering the mechanical properties at the contact interfaces. Critical mechanical properties at these contacts include normal stiffness, shear stiffness, friction, cohesion, tensile strength, and shear strength. Systematically investigating each of these parameters becomes imperative to comprehend their

collective impact on the elastic and strength properties of unconsolidated rocks under incremental confining pressure.

In this study, we analyzed the variation of elastic properties of unconsolidated sandstone with confining stress using a two-dimensional Particle Flow Code (PFC2D) (Itasca Inc, 2023). We employed the Linear Parallel Bond Model (LPBM) as the governing contact law (Potyondy & Cundall, 2004). Initially, we modeled granular rock structures resembling unconsolidated sandstones with Young's modulus ranging from 2-4 GPa, Poisson's ratio between 0.3-0.4, and unconfined compressive strength between 4-10 MPa using LPBM. Subsequently, we conducted numerical triaxial tests on digital rock samples at confining pressures ranging from 0 to 20 MPa. We then investigated how normal stiffness, shear stiffness, and contact area variations influenced the rocks' macroscopic behavior.

Two significant findings emerged from the DEM simulation of unconsolidated rocks: Firstly, maintaining a constant contact area at higher confining pressures may fail to replicate the compliance observed in natural rocks. Secondly, an elevation in the normal-to-shear stiffness ratio doesn't consistently enhance compliance in rocks under confining stress. In summary, the compliance of rocks under confining stress depends on both the ratio of grain contact normal to shear compliance and the grain contact area at a given confining pressure. These understandings are crucial in evaluating the shear failure nature of these rocks during wellbore washout, sanding issues and failure during increase in pore pressures.

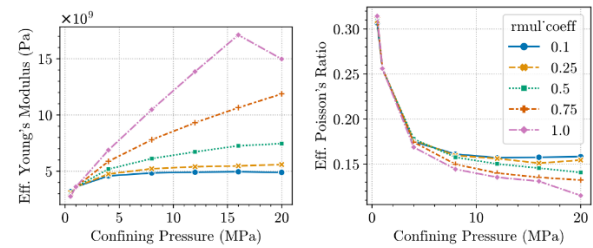


Figure 1: Variation of effective Young's modulus (right) and effective Poisson's ratio (left) with increase in confining pressures at different coefficient of radius multiplier power law in LPBM.

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Grain Contact Mechanics – DEM Simulation

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